Reading guide

Imbalanced semiprime representations of even integers

Slava Chan 10 October 2026

Current claim. For every sufficiently large even N, the draft claims at least $H_N/100000$ representations

$$N = p + rq, \qquad p, q \text{ prime}, \quad r = 1 \text{ or prime}, \quad r \le q^{0.74816964}.$$

Here $H_N = C_N N/\log^2 N$, with the singular factor in Section 1. Distinct primes p are counted once. The general claimed coefficient is $1.96176252 - 6.7617368\log(1/b)$ for fixed $1/2 < b < 1$. The secondary choice $b = 0.74816865$ gives $H_N/1000000$. No effective threshold for N is supplied.

A short route through the 82-page manuscript.

Result and normalization (Section 1)	p. 3
Restricted distribution and coefficient scope (Section 2)	p. 4
Positive upper sieve and full Chen weight (Sections 3–4)	pp. 7–10
Lower construction and finite evidence (Sections 5–6)	pp. 11–19
Bad-semiprime mass and exact transfer (Section 7)	p. 20
Comparison, limitations and AI disclosure (Section 8)	p. 21
Complete baseline modules (Appendices A–J)	from p. 25
Adopted same-mask refinement (Appendices K–L)	from p. 78

What changed. The revision supplies a restricted moving-residue argument for the coefficient families actually used, retaining the oscillatory phase and charging its loss. It uses a full eleven-term weight and paired geometric deletion. The old external 1.9728 coefficient and its derived branch are not inputs. A separate indexed library contains 151 supporting proof texts, with historical material clearly marked. The latest refinement uses $\omega(s) \ge 11/20$ for $s \ge 7/3$ on the same 17,572 paired cells; the 12/5 candidate is superseded.

Reproduce the finite checks. Use Python 3.11 or later, from the repository root; no third-party Python package is needed:

```
python verification/run_verification.py --work-dir replay-quick
```

The output directory must be new. The compact replay checks the input manifest, rational upper/lower ledgers, budgets and logarithm margins, and all 400 final entries after 430 rounds of 106 finite actions from the seed, then regenerates the adopted refinement using two interval implementations. The separate release archive supplies the complete selected numerical evidence and its inherited-binding audit. Consult each README for commands, hash manifests and the scope of recorded runs.

Review status. This is an AI-assisted research manuscript for mathematical scrutiny. Successful finite replays do not validate the whole analytic proof. The release does not claim fresh regeneration of every historical quadrature, complete machine formalization, or independent human peer review. Documented provenance edits remove private information; they do not alter mathematical coefficients or certificate arrays.